Solar Home Lighting System



Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **High** 2 labs

Covers these lab items: Home lighting system; Solar home lighting system

What it is used for

A solar home lighting system uses solar energy stored in a battery to power compatible lights and other approved loads. Operate the system within its rated capacity and follow the controller and battery instructions.

How to use

1. Inspect the solar panel, battery, controller, lights, cables, and connectors for damage.
2. Confirm that the system components and loads have compatible voltage ratings.
3. Ensure installation and wiring are performed by a qualified person using the approved design.
4. Check that the battery is connected with the correct polarity and controller sequence.
5. Position the solar panel securely where it receives adequate sunlight.
6. Switch on the system and verify that the lights and controller indicators operate normally.
7. Monitor battery status and avoid connecting loads beyond the system capacity.
8. Switch off the system safely before inspection or maintenance.

⚠ Safety

- Do not touch exposed conductors or damaged cables.
- Never short-circuit the battery or reverse its polarity.
- Use only compatible batteries and approved charging controllers.
- Keep the battery and electrical components dry and follow the required ventilation instructions.

Care & storage

- Clean the solar panel and check mounting, cables, and connectors regularly.
- Inspect the battery and controller according to the manufacturer's maintenance schedule.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.